# Evaluation runs: **fencing** (fencing-grove)

generated 2026-11-08 from brain-data 34d8e8fae 2026-11-08 19:23:20 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{ref}$, $G_{cand}$ = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{ref}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

No run of the stateless engine on this benchmark.

## stateful

No run of the stateful engine on this benchmark.

## offline

### offline: pair **fenced**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 164 | conforms | 0.4 | 8 | 10 min, 16 GB | 0.0 | 0.4 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 6 | 660 | conforms | 0.4 | 8 | 10 min, 16 GB | 0.1 | 0.4 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 20 | 2540 | conforms | 1.0 | 8 | 10 min, 16 GB | 0.0 | 1.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 70 | 9460 | conforms | 3.3 | 10 | 10 min, 16 GB | 0.1 | 3.3 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 252 | 35004 | conforms | 9.7 | 22 | 10 min, 16 GB | 0.1 | 9.7 | 0.1 | in-memory | brain05, 2026-11-02 |
| 6 | 924 | 130092 | conforms | 27.6 | 51 | 10 min, 16 GB | 0.2 | 27.2 | 0.4 | in-memory | brain05, 2026-11-02 |
| 7 | 3432 | 486972 | conforms | 60.2 | 180 | 10 min, 16 GB | 0.3 | 56.0 | 4.2 | in-memory | brain05, 2026-11-02 |
| 8 | 12870 | 1835700 | conforms | 233.6 | 735 | 10 min, 16 GB | 0.8 | 129.2 | 104.4 | in-memory | brain05, 2026-11-02 |
| 9 | 48620 | 6961916 | 🕓 (1 h) | | | 1 h, 250 GB | 1.7 | 588.8 | | in-memory | brain05, 2026-11-03 |

### offline: pair **stale-frontend**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 176 | 3 | violation | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 6 | 1152 | 40 | violation | 0.3 | 8 | 10 min, 16 GB | 0.1 | 0.3 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 20 | 12128 | 318 | violation | 2.4 | 10 | 10 min, 16 GB | 0.0 | 2.4 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 70 | 158080 | 2532 | violation | 8.4 | 22 | 10 min, 16 GB | 0.1 | 8.3 | 0.1 | in-memory | brain05, 2026-11-02 |
| 5 | 252 | 2230176 | 20578 | violation | 68.7 | 147 | 10 min, 16 GB | 0.1 | 67.5 | 1.2 | in-memory | brain05, 2026-11-02 |
| 6 | 924 | 32497920 | | violation | 1196.1 | 1305 | 10 min, 16 GB | 0.2 | 1169.0 | 27.1 | in-memory | brain05, 2026-11-02 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 6 stale-frontend fifo, 7 stale-frontend fifo, 7 fenced fifo, 11 fenced fifo.

skipped: a smaller size had reached its limit: 7 stale-frontend fifo, 8 stale-frontend fifo, 9 stale-frontend fifo.

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **fenced**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 164 | conforms | 5.3 | 194 | 10 min, 16 GB | 0.0 | 5.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 6 | 660 | conforms | 11.7 | 6316 | 10 min, 16 GB | 0.1 | 11.7 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 20 | 2540 | conforms | 40.8 | 8064 | 10 min, 16 GB | 0.0 | 40.8 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Java implementation against the Rust specification: pair **stale-frontend**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 176 | 3 | violation | 4.5 | 203 | 10 min, 16 GB | 0.0 | 4.5 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 6 | 1152 | 40 | violation | 11.7 | 5863 | 10 min, 16 GB | 0.1 | 11.7 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 20 | 12128 | 318 | violation | 125.4 | 8170 | 10 min, 16 GB | 0.0 | 125.3 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (cross offline, Java implementation against the Rust specification).** planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **fenced**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 164 | conforms | 3.6 | 166 | 10 min, 16 GB | 3.6 | 0.4 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 6 | 660 | conforms | 3.8 | 166 | 10 min, 16 GB | 3.8 | 0.4 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 20 | 2540 | conforms | 3.8 | 161 | 10 min, 16 GB | 3.8 | 1.0 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Rust implementation against the Java specification: pair **stale-frontend**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 176 | 3 | violation | 3.6 | 166 | 10 min, 16 GB | 3.6 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 6 | 1152 | 40 | violation | 3.8 | 166 | 10 min, 16 GB | 3.8 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 20 | 12128 | 318 | violation | 3.8 | 163 | 10 min, 16 GB | 3.8 | 2.4 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (cross offline, Rust implementation against the Java specification).** planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **fenced**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 164 | conforms | 3.6 | 162 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 2 | 6 | 660 | conforms | 4.0 | 162 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 20 | 2540 | conforms | 4.8 | 157 | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **stale-frontend**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 176 | 60 | violation | 3.6 | 162 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 2 | 6 | 1152 | 828 | violation | 4.1 | 162 | 1 h, 250 GB | 6 | brain01, 2026-11-02 |
| 3 | 20 | 12128 | 11076 | violation | 6.2 | 157 | 1 h, 250 GB | 19 | brain01, 2026-11-02 |

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **fenced**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 164 | 164 | conforms | 5.3 | 194 | 10 min, 16 GB | 5.3 | 0.4 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 660 | 660 | conforms | 11.8 | 6316 | 10 min, 16 GB | 11.7 | 0.4 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 2540 | 2540 | conforms | 40.9 | 8064 | 10 min, 16 GB | 40.8 | 1.0 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Rust implementation against its Java twin: pair **stale-frontend**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 176 | 176 | conforms | 4.5 | 203 | 10 min, 16 GB | 4.5 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 1152 | 1152 | conforms | 11.7 | 5863 | 10 min, 16 GB | 11.7 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 12128 | 12128 | conforms | 125.4 | 8172 | 10 min, 16 GB | 125.3 | 2.4 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (twins offline, Rust implementation against its Java twin).** planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

## twins offline, Java implementation against its Rust twin

### twins offline, Java implementation against its Rust twin: pair **fenced**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 164 | 164 | conforms | 5.3 | 194 | 10 min, 16 GB | 0.4 | 5.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 660 | 660 | conforms | 11.7 | 6316 | 10 min, 16 GB | 0.4 | 11.7 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 2540 | 2540 | conforms | 40.8 | 8064 | 10 min, 16 GB | 1.0 | 40.8 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Java implementation against its Rust twin: pair **stale-frontend**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 176 | 176 | conforms | 4.5 | 203 | 10 min, 16 GB | 0.1 | 4.5 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 1152 | 1152 | conforms | 11.7 | 5863 | 10 min, 16 GB | 0.3 | 11.7 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 12128 | 12128 | conforms | 125.4 | 8172 | 10 min, 16 GB | 2.4 | 125.3 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (twins offline, Java implementation against its Rust twin).** planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **fenced**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 164 | 164 | conforms | 5.0 | 239 | 10 min, 16 GB | 5.0 | 0.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 2 | 660 | 660 | conforms | 10.1 | 949 | 10 min, 16 GB | 10.1 | 0.3 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 2540 | 2540 | conforms | 38.1 | 7634 | 10 min, 16 GB | 38.1 | 1.2 | 0.0 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **stale-frontend**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 176 | 176 | conforms | 4.6 | 204 | 10 min, 16 GB | 4.6 | 0.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 2 | 1152 | 1152 | conforms | 12.1 | 5870 | 10 min, 16 GB | 11.9 | 0.4 | 0.2 | in-memory | brain02, 2026-11-03 |
| 3 | 12128 | 12128 | conforms | 109.2 | 7833 | 10 min, 16 GB | 109.2 | 2.5 | 0.0 | in-memory | brain02, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **fenced**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 164 | 164 | conforms | 5.0 | 239 | 10 min, 16 GB | 0.1 | 5.0 | 0.0 | in-memory | brain02, 2026-11-03 |
| 2 | 660 | 660 | conforms | 10.1 | 949 | 10 min, 16 GB | 0.3 | 10.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 2540 | 2540 | conforms | 38.1 | 7634 | 10 min, 16 GB | 1.2 | 38.1 | 0.0 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **stale-frontend**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 176 | 176 | conforms | 4.6 | 204 | 10 min, 16 GB | 0.1 | 4.6 | 0.0 | in-memory | brain02, 2026-11-03 |
| 2 | 1152 | 1152 | conforms | 12.1 | 5870 | 10 min, 16 GB | 0.4 | 11.9 | 0.1 | in-memory | brain02, 2026-11-03 |
| 3 | 12128 | 12128 | conforms | 109.2 | 7833 | 10 min, 16 GB | 2.5 | 109.2 | 0.0 | in-memory | brain02, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **fenced**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 164 | 164 | conforms | 5.4 | 170 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2 | 660 | 660 | conforms | 10.6 | 942 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 2540 | 2540 | conforms | 42.7 | 8067 | 1 h, 250 GB | | brain02, 2026-11-03 |

**context-free twins online, Java port (JMC) against the Rust implementation's export: pair stale-frontend, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 176 | 176 | conforms | 4.7 | 211 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2 | 1152 | 1152 | conforms | 13.0 | 5610 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 12128 | 12128 | conforms | 129.3 | 8143 | 1 h, 250 GB | | brain02, 2026-11-03 |

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair fenced, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 164 | 164 | conforms | 5.7 | 235 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2 | 660 | 660 | conforms | 10.4 | 945 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 2540 | 2540 | conforms | 39.2 | 7630 | 1 h, 250 GB | | brain02, 2026-11-03 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair stale-frontend, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 176 | 176 | conforms | 4.6 | 200 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2 | 1152 | 1152 | conforms | 12.3 | 5866 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 12128 | 12128 | conforms | 111.9 | 7829 | 1 h, 250 GB | | brain02, 2026-11-03 |